Tangzheng Lian

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Education

King's College London *London, UK*
PhD Candidate in Computer Science *Jun 2023–Present*
- Supervisor: Dr. Oya Celiktutan
- Funding: The Natural, Mathematical & Engineering Sciences (NMES) Faculty Studentship

University of Birmingham *Birmingham, UK*
MSc in Computer Science ***(Distinction)*** *Sept 2021–Dec 2022*

Beijing University of Posts and Telecommunications *Beijing, China*
BEng in Telecommunication Engineering *Sept 2017–Jun 2021*

Research Interests

Trustworthy AI/ML: Algorithmic Biases/Fairness; Out-of-Distribution (OOD)/Domain Generalization

Multimodal Learning: Vision–Language Models (VLMs); Multimodal Large Language Models (MLLMs)

Generative Modelling: Text-to-Image and Text-to-Motion Generation; Diffusion Models; Flow Matching

Affective Computing: Facial Behaviour Understanding; Gaze Communication; Speech Recognition

Publications

Published

[1] **Tangzheng Lian**, Guanyu Hu, Yijing Ren, Dimitrios Kollias, Oya Celiktutan. *A Closed-Form Solution for Debiasing Vision-Language Models with Utility Guarantees Across Modalities and Tasks*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

[2] **Tangzheng Lian**, Guanyu Hu, Dimitrios Kollias, Xinyu Yang, Oya Celiktutan. *Fair Domain Generalization: An Information-Theoretic View*. In Proceedings of the AAAI Conference on Artificial Intelligence ***(Oral)***, 2026.

[3] Guanhe Huang, Songqiao Han, **Tangzheng Lian**, Oya Celiktutan. *Improving the Diffusability of Motion Tokenizer*. Accepted at the Conference on Neural Information Processing Systems (NeurIPS, ***Spotlight***), 2026.

[4] Guanyu Hu, **Tangzheng Lian**, Dimitrios Kollias, Oya Celiktutan, Xinyu Yang. *CausalAffect: Causally Guided Learning of Psychology-Aligned Facial Affect Relations*. Accepted at the Conference on Neural Information Processing Systems (NeurIPS), 2026.

[5] Guanyu Hu, **Tangzheng Lian**, Na Yan, Dimitrios Kollias, Xinyu Yang, Oya Celiktutan, Siyang Song, Zeyu Fu. *FairMT: Fairness for Heterogeneous Multi-Task Learning*. Accepted at the Conference on Neural Information Processing Systems (NeurIPS), 2026.

[6] **Tangzheng Lian**, Oya Celiktutan. *A Feature-level Framework for Evaluating Demographic Biases in Facial Expression Recognition Models*. IEEE Transactions on Affective Computing, 2026.

[7] **Tangzheng Lian**, Oya Celiktutan. *Towards Pareto Efficiency in Fair Facial Expression and Action Unit Recognition*. In Proceedings of the IEEE International Conference on Automatic Face and Gesture Recognition (FG), 2026.

[8] Cheng Peng, **Tangzheng Lian**, Oya Celiktutan. *When Gaze Meets Speech: Multimodal Multi-Person Speech and Gaze Behaviour Understanding*. Accepted to the British Machine Vision Conference (BMVC), 2026.

[9] **Tangzheng Lian**, David Adama, Pedro Machado, Doratha Vinkemeier. *Supervised Contrastive Learning with Identity-Label Embeddings for Facial Action Unit Recognition*. In Proceedings of the British Machine Vision Conference (BMVC), 2023.

[10] Wenqi Xu, **Tangzheng Lian**, Wei Liu, Kaili Zhao. *Attention-guided Soft Ranking Loss for Resource-constrained Head Pose Estimation*. In Proceedings of the IEEE International Conference on Network Intelligence and Digital Content, 2021.

Under Review

[11] **Tangzheng Lian**, Yijing Ren, Guanhe Huang, Oya Celiktutan. *Fair Image Classifiers Under Unknown Distribution Shifts*. Submitted to the AAAI Conference on Artificial Intelligence, 2026.

Work/Research Experience

Huawei Noah's Ark Lab *London, UK*

Research Scientist Intern at Darwin Research Center *Aug 2026–Present*

- Working on on-policy self-distillation for fine-grained multimodal perception using MLLMs (Qwen 3.5 series).
- Developed a gaze-aware wearable AI assistant integrating Pupil Labs Neon glasses with Whisper, Gemini 3.5 Flash Lite, and GPT-5.4 mini via OpenRouter, enabling real-time streaming audio-visual understanding in the user's visual focus.
- Collected a multimodal dataset for gaze and attention analysis. combining 3.1 million gaze samples with scene videos, 3D eye states, IMU signals, and eye-movement events.

Honda R&D UK Ltd *Reading, UK*

Associated Researcher *Jun 2026–Aug 2026*

- Collected a novel multimodal benchmark for collaborative human-human-object interaction, synchronizing rigid-object tracking, multi-camera full-body motion capture, RGB-D and egocentric video, and Wuji glove-based hand-pose/tactile sensing to support motion generation, intention understanding, and hand-object interaction in human-robot collaboration scenarios.

Queen Mary University of London *London, UK*

Visiting PhD *Nov 2024–Oct 2025*

- Developed a weakly supervised, causally guided graph-learning framework to model psychology-aligned AU-to-AU and AU-to-expression relations, improving Action Unit detection and facial expression recognition across six benchmarks [4].
- Developed a unified fairness framework for heterogeneous multi-task learning across binary, multi-class classification, and regression through fairness coordinates, water-filling allocation, and head-aware optimization while preserving utility [5].

King's College London *London, UK*

PhD Candidate in the Department of Engineering *Jun 2023–Present*

- Developed a training- and data-free debiasing method for VLMs with a closed-form solution and utility guarantees in the cross-modal embedding space, outperforming prior debiasing methods in group and intersectional fairness across downstream tasks including zero-shot image classification, text-to-image retrieval, and generation [1].
- Addressed the problem of fair domain generalization by deriving information-theoretic bounds for both expected risk and fairness violations under domain shift, and developing a Pareto optimization method that outperforms SOTA methods with improved utility–fairness trade-offs on vision and language datasets [2].
- Evaluated algorithmic biases in facial expression recognition (FER) models by introducing a feature-level fairness evaluation framework, and developed a multi-objective optimization approach to characterize the utility–fairness trade-off across FER, facial action unit recognition, and facial attribute classification [6, 7].
- Developed the Diffusable Motion Tokenizer (DiMoT) framework, combining diffusion-guided Critic Spectral Adaptation to suppress high-frequency latent artifacts with an Alpha-Flow Motion Decoder for one-step generative decoding, achieving state-of-the-art text-to-motion generation in fidelity and text–motion alignment [3].
- Introduced EasyCom-Gaze, an egocentric benchmark with synchronized gaze and speaker annotations, and developed a multimodal teacher–student framework that jointly models multi-person gaze and speech with MLLM semantic guidance at scene level, achieving state-of-the-art performance on both tasks [8].

Nottingham Trent University *Nottingham, UK*

Research Assistant at the Computing and Informatics Research Center (CIRC) *Apr 2022–Aug 2023*

- Addressed poor generalization to unseen subjects and the challenge of modelling correlations among facial action units (AUs) by proposing a novel AU detection framework that achieved SOTA performance on the BP4D and DISFA benchmarks [9].
- Annotated video sequences of behaviours and activities of autistic primary school students in Nottingham, enabling automatic behavioural analysis for autism research in real-world educational environments as part of the AI-TOP project.

Awards, Studentships, & Grants

NMES PhD studentship (2023), King's College London *£25,600 p.a. & tuition waiver*

The CIRC Strategic Research Funding (2022), Nottingham Trent University *£32,700*

The Outstanding Undergraduate Thesis (2021), Beijing Municipal Education Commission

Presentations and Talks

Oral presentation at the AAAI Conference (Main Technical Track), Singapore, January 2026. Available online: Website

Invited talk at the BMVA Symposium on AI Security, London, July 2025. Available online: Website, YouTube.

Teaching Experience

Sensing and Perception (Spring 2024), Teaching Assistant — *King's College London*
Engineering Mathematics (Fall 2023), Teaching Assistant — *King's College London*
Applied Maths and Graphics (Spring 2023), Teaching Assistant — *Nottingham Trent University*

Professional Services

Professional Memberships: BMVA, IAPR, AAAI.

Conference Reviewer: CVPR, AAAI, WACV, BMVC.

Journal Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Affective Computing, IEEE Transactions on Big Data.

Skills

- **Programming Languages:** Python
- **ML Frameworks:** PyTorch, Hugging Face, NumPy, Pandas
- **Tools:** Git, Linux, Slurm-based HPC (B/H200 clusters), Anaconda, LaTeX
- **Software:** VS Code, OptiTrack, Motive, Captury, Pupil Labs SDK, Wuji Glove SDK.